

# Stream Error Flags in C++

## 1. eofbit

### Meaning:

Set when the stream tries to read past the end of file (EOF).

**When does it happen?** - File ka aakhri character read ho chuka ho - Next read operation fail ho jaye because file end is reached

### Example:

```
ifstream in("data.txt");
char ch;

while (in.get(ch)) {}    // read until end

if (in.eof()) {
    cout << "EOF reached!";
}
```

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## 2. failbit

### Meaning:

Set when a logical error occurs — usually wrong input format.

**When does it happen?** - User gives wrong type (e.g., text instead of number) - File open fails - Searching/reading format mismatch

### Example:

```
int x;
cin >> x;    // user enters: ABC

if (cin.fail()) {
    cout << "Logical error: failbit set!";
}
```

### Another example:

```
ifstream in("no_such_file.txt");
if (!in) {
    cout << "File open failed (failbit set)";
}
```

### 3. badbit

**Meaning:**

Set when a serious low-level I/O error occurs.

**When does it happen?** - Hardware issue - Disk error - Stream buffer corruption - Illegal operation on buffer

**Example:**

```
fstream fs;
fs.rdbuf()->sputc EOF; // force a low-level error

if (fs.bad()) {
    cout << "Low-level I/O error: badbit set!";
}
```

## ★ Summary Table

| Flag    | When It Is Set              | Type of Error |
|---------|-----------------------------|---------------|
| eofbit  | Read hits end-of-file       | Mild error    |
| failbit | Format error / read failure | Logical error |
| badbit  | Stream/internal I/O damage  | Serious error |

## ★ One-Line Definitions (Exam-Perfect)

- **eofbit:** Set when input operation reaches end of file.
- **failbit:** Set when an operation logically fails (wrong input).
- **badbit:** Set when a fatal internal error occurs (I/O damage).